INTERNAL DOCUMENT

NiceOS

Market Activation & Intelligence Platform

A Systems Proposal for Market Link Ltd — Nice Millers Ltd Engagement

Prepared for:

Market Link Ltd — Management

(Engagement client: Nice Millers Ltd)

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01 EXECUTIVE SUMMARY

This is an internal systems proposal prepared for Market Link Ltd's management. It outlines NiceOS, a Market Activation & Intelligence Platform to be built for Market Link's field marketing operations under its engagement with Nice Millers Ltd. Market Link owns the client relationship and reporting responsibility with Nice Millers; NiceOS is the internal system Market Link's own field team will use to plan routes, activate retail outlets, and generate the market intelligence Market Link reports back to its client.

NiceOS equips field sales representatives with an offline-first mobile app for route-guided visits and structured data capture, and gives Market Link management a live, verified view of market coverage, retailer health, and competitor activity — turning field activity into decision-ready intelligence that strengthens both internal operations and the Nice Millers account relationship.

This proposal is scoped specifically to market activation, route planning, field data collection, and market analytics for the Nice Millers engagement. Milling/production operations, inventory management, and HR & payroll sit with Nice Millers directly and are intentionally out of scope for this build.

02 OUR UNDERSTANDING OF THE ENGAGEMENT

Market Link Ltd has been engaged by Nice Millers Ltd to grow and manage its retail market presence — mapping territories, activating retail outlets, and running field sales activity on Nice Millers' behalf. Running an engagement of this kind without dedicated tooling creates real operational risk for Market Link:

- No Territory Visibility — retailers and prospects exist informally; there is no structured, mapped registry of who is being served and who isn't.
- Unverified Field Activity — without GPS or timestamped check-ins, it's difficult to confirm whether a rep actually visited a retailer or simply reported that they did.
- No Structured Route Planning — reps plan their own days with no system-guided prioritisation of high-value or overdue outlets.
- Blind to Competitor Activity — no consolidated record of what rival brands are doing on the shelves Nice Millers products compete on.
- No Early Warning on Churn — a retailer that quietly stops ordering is discovered late, if at all, instead of being flagged automatically.
- No Reporting Layer for Nice Millers — Market Link has no live, evidence-backed way to show its client the value the engagement is delivering.

03 PROPOSED SOLUTION — NICEOS

NiceOS is a cloud-based Market Activation & Intelligence Platform built for Market Link's field operations on the Nice Millers account. It is role-aware (reps, managers, and directors each see what they need), offline-first for field reliability, and designed to turn every field visit into structured, verified data.

M1 Territory & Retailer Mapping

Know exactly who is out there, and where.

- Retailer/duka/distributor registry — name, owner, contact, GPS location, business type & size
- Status tagging — active customer, prospect, inactive, or competitor-only outlet
- Territory hierarchy — Region → County → Subcounty → Ward → Sales Territory
- Each territory assigned to a rep with defined targets and visit frequency

M2 Route Planning & Optimisation

Every rep starts the day with a plan, not a guess.

- System-generated daily routes based on visit priority, frequency due, and geography
- Manager review and adjustment before rollout
- Shortest-path sequencing to reduce travel time between outlets

M3 Field Data Collection (Mobile App)

Offline-first — built for real connectivity conditions.

- GPS- and timestamp-verified check-in at every outlet — no fake or phantom visits
- Shelf & stock capture — photos, stock levels, shelf facing, expiry
- Competitor intelligence — rival brand pricing, promotions, and shelf presence per outlet
- Order-intent capture — reps log what a retailer wants to (re)order
- Order-intent automatically forwarded to the office via WhatsApp/email for follow-up
- Capture → store locally → sync automatically once connectivity returns

M4 Retailer Intelligence & Market Analytics

Field data becomes market intelligence.

- Per-retailer profile — visit history, order pattern, competitor exposure
- Retailer health score — active, growing, declining, or lost, inferred from visit and order cadence
- Churn-risk flagging — e.g. an outlet that used to reorder every 9 days and has gone silent for 22
- Coverage maps — which wards/territories are visited, under-visited, or untouched
- Opportunity engine — high-outlet-density areas with low current penetration
- Competitor heatmap — where rival brands are winning shelf space
- Rep KPI dashboard — visit count, strike rate, distance covered, new outlets onboarded

Platform-Wide Capabilities

- Offline-first architecture — the mobile app is fully usable with zero connectivity; data syncs automatically once a signal is available
- Role-based access control — Admin, Territory Manager, Sales Rep, and Client Viewer roles, each scoped to what they need
- Full audit trail — every visit, capture, and order-intent entry logged with user, location, and timestamp
- WhatsApp/email forwarding — order-intent and key alerts routed to the office automatically
- Director/HQ dashboard — live coverage, rep performance, and market intelligence in one view
- Mobile-responsive web admin — territory, retailer, and rep management from any device

04 USER ROLES & RESPONSIBILITIES

NiceOS is only as useful as the people using it. This chapter breaks down every role in the system — what each person does day to day, exactly which data they touch, and what part of the enterprise benefits as a result. This is the clearest way to see the value chain the platform creates, from a single rep's visit to Market Link's reporting relationship with Nice Millers.

R1 Market Link Platform Admin

Owns the system end-to-end.

WHAT THEY DO

- Create and structure territories, regions, and sales routes
- Assign reps to territories and set visit targets
- Approve or adjust system-generated routes
- Monitor performance across all territories and reps
- Manage user accounts and role permissions
- View full, unfiltered market intelligence and analytics

DATA THEY INTERACT WITH

- Full system data — every territory, retailer, rep, and visit record
- Configuration data — targets, territory boundaries, user accounts
- Aggregated analytics across the entire Nice Millers account

VALUE TO THE ENTERPRISE *Total operational control without needing developer involvement to reconfigure how the account is run.*

R2 Territory / Regional Manager

Owns one region, and is accountable for it.

WHAT THEY DO

- Monitor reps assigned to their territory
- Review and approve daily visit routes
- Track sales and coverage trends within their region
- Compare performance across territories they oversee
- Escalate churn-risk retailers and competitor activity

DATA THEY INTERACT WITH

- Visit records and KPIs for reps in their assigned territory
- Retailer health scores and churn alerts within their region
- Route plans awaiting approval

VALUE TO THE ENTERPRISE *Real-time regional management instead of waiting on manually compiled field reports.*

R3 Field Sales Representative

The most important user — this is where the data comes from.

WHAT THEY DO

- Log in each morning and receive the day's route
- Navigate to each assigned retailer in sequence
- Check in with GPS verification on arrival
- Capture stock levels, shelf photos, and competitor data
- Log order-intent from the retailer
- Leave visit notes and move to the next stop

DATA THEY INTERACT WITH

- Their own assigned route and retailer profiles for the day
- Write access to visit records, stock/shelf/competitor data, and order-intent
- Their own performance history and targets

VALUE TO THE ENTERPRISE *A guided, structured workday with less wasted travel and an objective record of their own performance.*

R4 Nice Millers Account Viewer

Read-only window into the engagement, for the client.

WHAT THEY DO

- View coverage maps and territory performance for their brand
- View aggregated KPIs — visits, growth, retailer health trends
- View market intelligence summaries — competitor activity, demand signals
- No edit access — cannot see internal rep management, targets, or Market Link's operational detail

DATA THEY INTERACT WITH

- Aggregated, read-only dashboards scoped to the Nice Millers account
- Coverage %, retailer growth trend, order-intent volume trend
- Competitor intelligence summary

VALUE TO THE ENTERPRISE *Nice Millers gets evidence-backed confidence in what the engagement is delivering, without Market Link exposing its internal playbook.*

05 DATA INTELLIGENCE FRAMEWORK

Every data point a rep captures in the field exists to feed a specific piece of market intelligence — nothing is collected without a defined purpose. The table below traces that path: from what the rep taps or photographs, to the metric it produces, to where that metric shows up for managers and Nice Millers.

Data Point Captured	How It's Captured	Intelligence Metric It Feeds	Surfaces In
GPS check-in + timestamp	Location and time logged automatically on arrival at an outlet	Verified Visit Rate — % of visits confirmed by GPS, not self-reported	Rep KPI Dashboard
Check-in duration	Time between check-in and check-out at the outlet	Average Visit Duration — flags rushed or padded visits	Rep KPI Dashboard
Visit outcome / notes	Rep records outcome — completed, no stock, not interested, closed	Visit Outcome Distribution — recurring objections and blockers	Territory Reports
Stock level per SKU	Rep logs current stock observed at the outlet	Stockout Risk & Reorder Signal	Retailer Profile, Opportunity Engine
Shelf facing / photo	Rep photographs shelf space and product facing	Shelf Share Index — Nice Millers' visual presence vs. competitors	Retailer Profile, Brand Visibility Report
Product expiry data	Rep notes expiry dates observed on shelf	Expiry Risk Alert	Executive Alerts
Competitor product present (Y/N + brand)	Rep flags rival products stocked at the outlet	Competitor Presence Map	Territory Map, Competitor Heatmap
Competitor price observed	Rep logs rival product pricing at the outlet	Price Gap Analysis — Nice Millers pricing vs. market	Market Intelligence Report
Competitor promotion observed	Rep notes active rival promotions at the outlet	Competitive Activity Alert	Executive Alerts
Order-intent (SKU + quantity)	Rep logs what the retailer wants to (re)order	Demand Pipeline / Order Conversion Rate	WhatsApp/Email Forward, Demand Report
Outlet status change	Rep or system updates status — prospect → active → inactive	Retailer Lifecycle Stage	Retailer Health Score
Visit cadence vs. planned frequency	System compares actual visit dates to the retailer's target frequency	Churn Risk Score — flags retailers going quiet	Retailer Profile, Executive Alerts
New outlet registration	Rep captures a new, previously unmapped retailer with GPS + profile	Market Penetration Growth	Opportunity Engine, Coverage Map
Rep movement between stops (GPS trail)	Background location logging while the app is active	Route Efficiency / Distance Travelled	Rep KPI Dashboard

This is the core discipline behind NiceOS: every field of every form has a downstream destination. If a data point can't be traced to a metric on this list, it doesn't belong in the mobile app.

06 DASHBOARD & SCREEN CONCEPTS

The wireframes below are structural only — they show layout and where each type of data will sit on screen, not final visual design, colours, or branding. They exist to validate the information architecture before any UI design work begins.

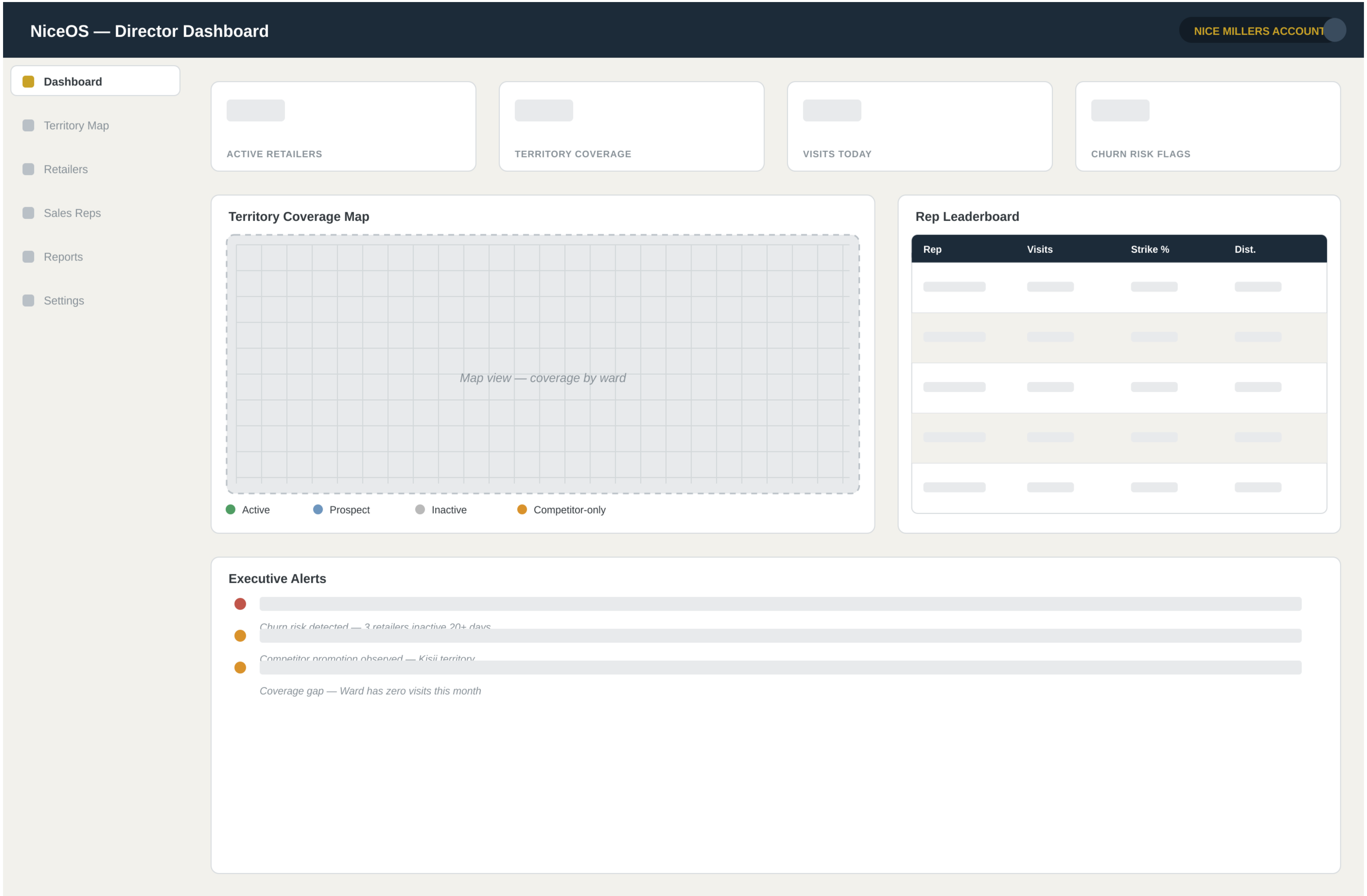

Figure 1 — Director Dashboard (Web): KPI cards, territory coverage map, rep leaderboard, and executive alerts in one view.

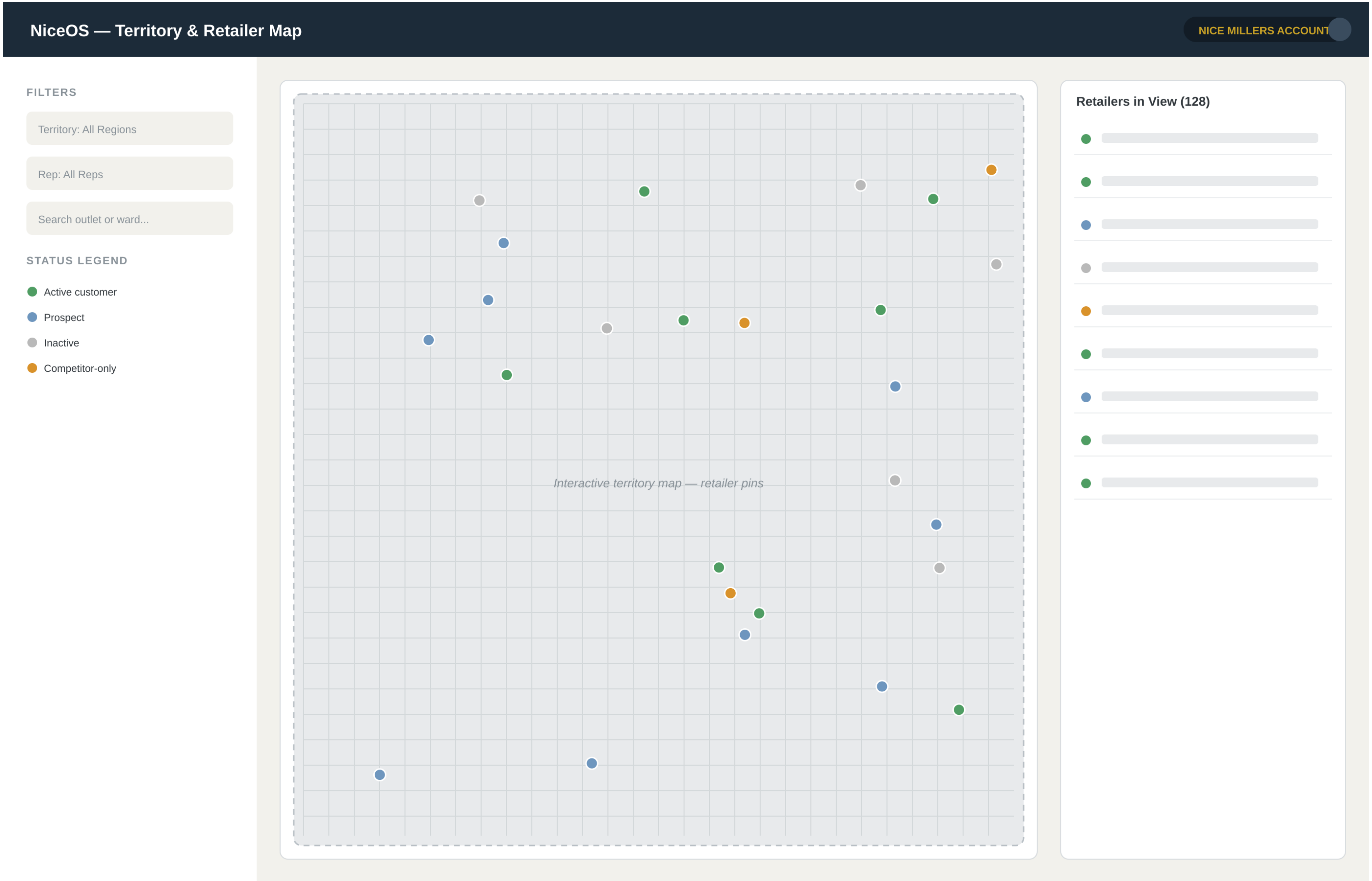

Figure 2 — Territory & Retailer Map (Web): filterable map of all outlets by status, with a live retailer list.

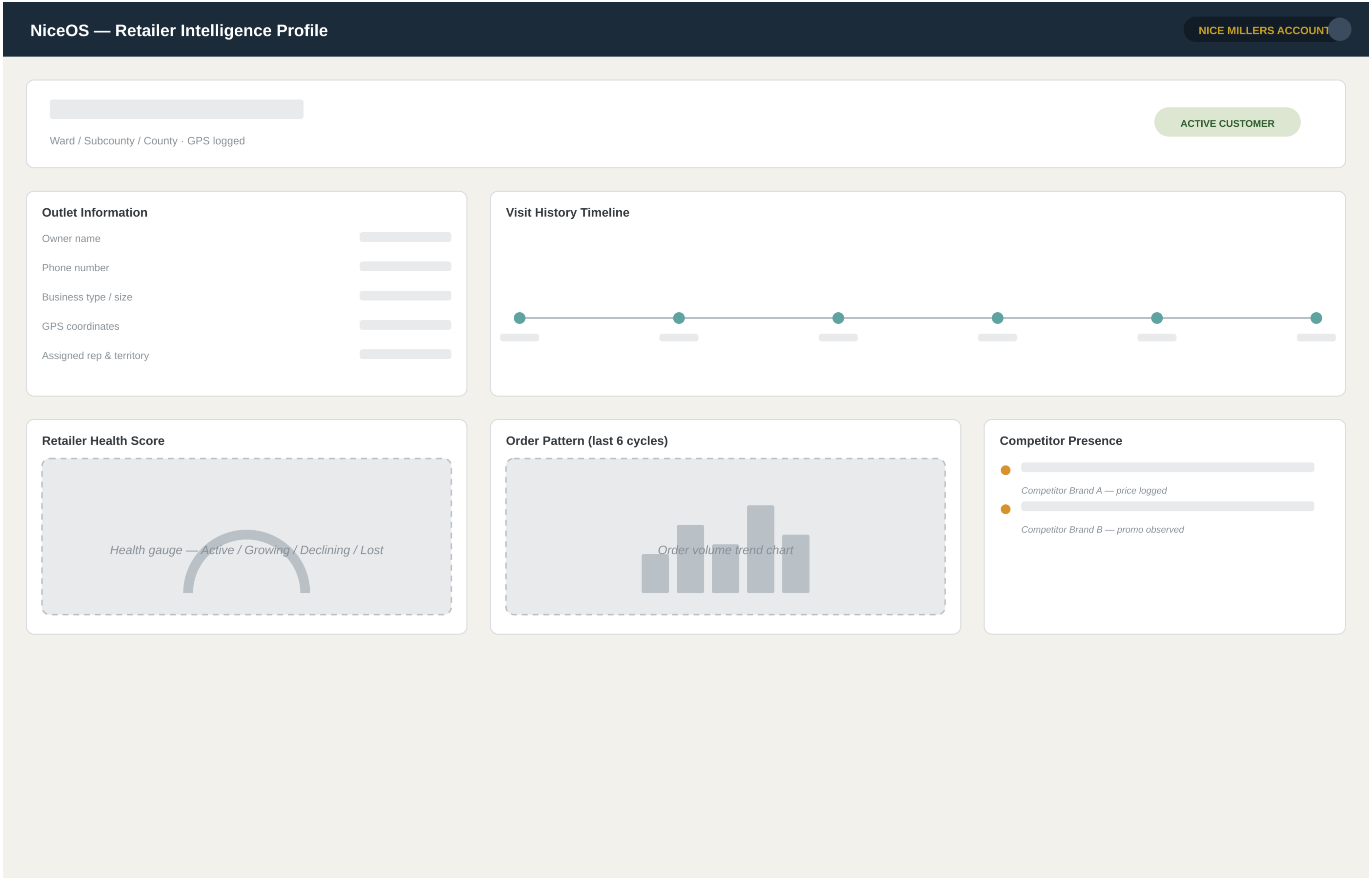
NiceOS — Retailer Intelligence Profile
NICE MILLERS ACCOUNT
Ward / Subcounty / County · GPS logged
ACTIVE CUSTOMER
Outlet Information
Owner name
Phone number
Business type / size
GPS coordinates
Assigned rep & territory
Visit History Timeline
Retailer Health Score
Health gauge — Active / Growing / Declining / Lost
Order Pattern (last 6 cycles)
Order volume trend chart
Competitor Presence
Competitor Brand A — price logged
Competitor Brand B — promo observed

Figure 3 — Retailer Intelligence Profile (Web): outlet info, health score, visit timeline, order pattern, and competitor presence.

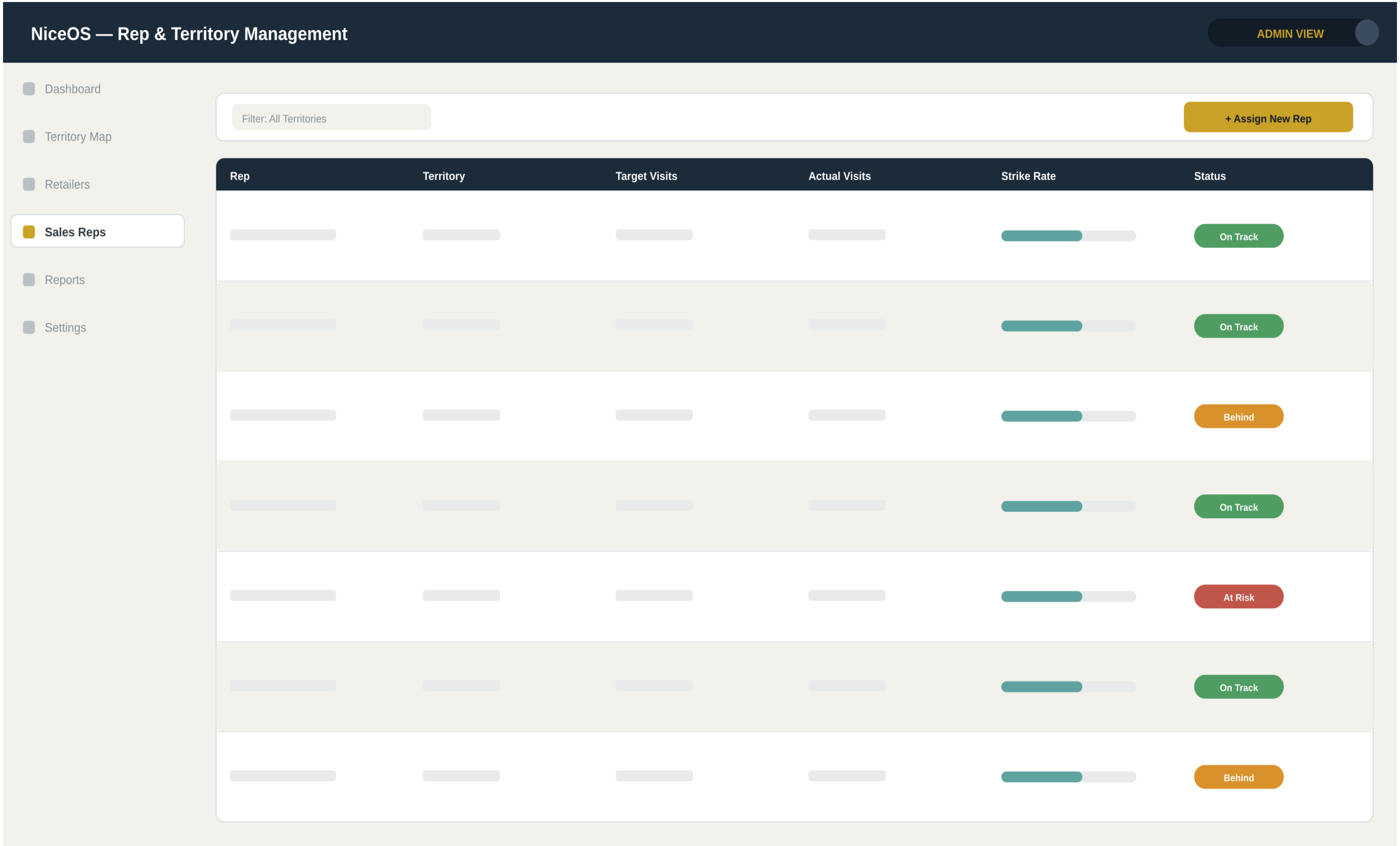

Figure 4 — Rep & Territory Management (Web, Admin): target vs. actual visits and status per rep.

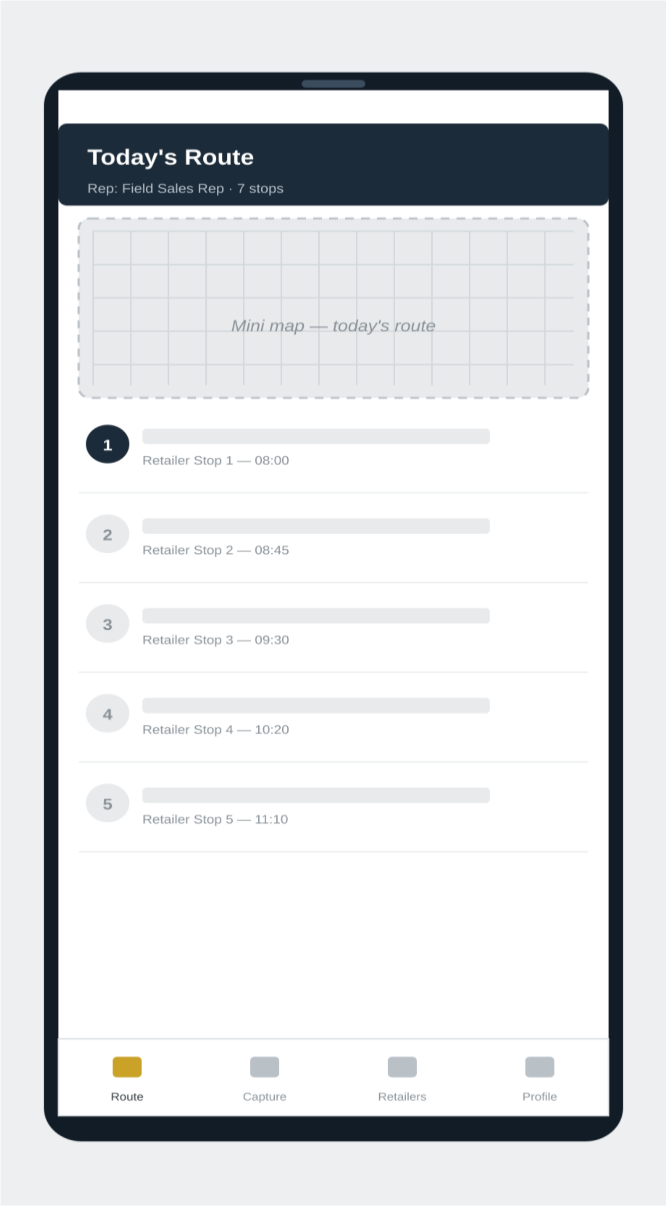

Figure 5 — Mobile: Today's Route

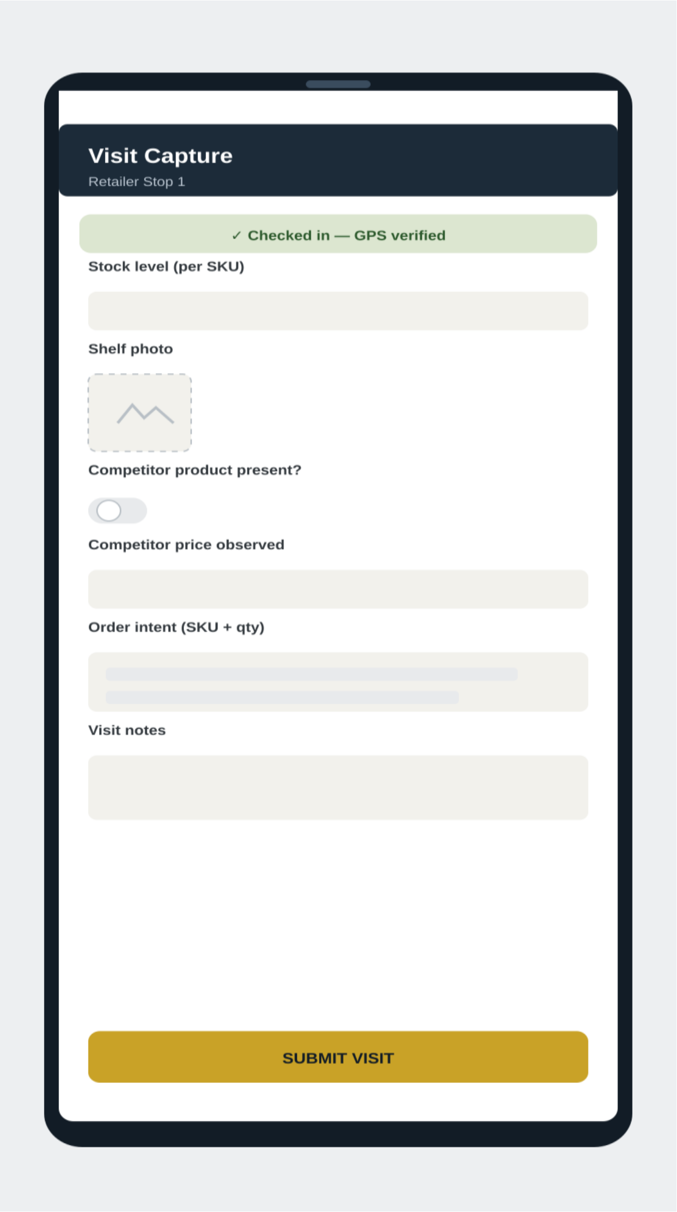

Figure 6 — Mobile: Visit Capture

07 HOW NICEOS ADDS VALUE

Concrete operational outcomes from each part of the platform:

Verified Field Presence	GPS- and time-verified check-ins eliminate phantom visits and reporting guesswork, so management can trust what the field is telling them.
Coverage Visibility	A live map shows exactly which wards and territories are covered, under-served, or untouched — no more guessing where the gaps are.
Demand Signal Capture	Order-intent logged by reps becomes a structured demand pipeline instead of scattered notes or memory, forwarded straight to the office.
Competitor Awareness	Real-time visibility into rival brand pricing, promotions, and shelf presence, outlet by outlet.
Faster Decisions	Market Link management gets a live, consolidated view of the market instead of waiting on manually compiled field reports.
Early Churn Detection	Retailers going quiet are flagged automatically, before the account is lost for good.
Stronger Client Reporting	Nice Millers sees evidence-backed engagement results through their own read-only view, strengthening the account relationship.

08 TECHNOLOGY & DELIVERY APPROACH

Technology Stack

- Web Admin & Dashboard — Next.js (App Router) + Tailwind CSS, deployed on Vercel
- Database — Supabase / PostgreSQL with PostGIS for territory, route, and retailer geo-data
- Mobile App — Flutter (Android-first), offline storage with local sync queue, background sync on reconnect
- Notifications — WhatsApp Business Cloud API and email for order-intent and alert forwarding
- Mapping — PostGIS-backed spatial queries with a web mapping layer (MapLibre/Leaflet) for coverage and heatmaps
- Security — role-based access, encrypted data in transit and at rest, full audit logging

Delivery Timeline

An agile delivery model with biweekly internal demos to Market Link. Estimated delivery: approximately 14 days from kick-off.

Phase	Deliverable	Duration
Phase 1	Discovery & Architecture — territory model, data schema, cloud infrastructure setup	1 day
Phase 2	Core Platform — auth, roles, territory & retailer mapping, admin dashboard shell	2 days
Phase 3	Mobile App — offline-first field data collection, route display, GPS check-in, sync engine	4 days
Phase 4	Retailer Intelligence & Analytics — health scores, churn flags, coverage maps, competitor reporting	3 days
Phase 5	Integration & Deployment — WhatsApp/email forwarding, data migration, training, go-live	4 days

09 DEVELOPMENT INFRASTRUCTURE REQUIREMENTS

Delivering an offline-first mobile app on top of a geo-heavy data model (PostGIS, mapping, route calculation) alongside a live web dashboard is a heavier development workload than a typical CRUD app. To deliver NiceOS on schedule and at quality, the following development environment is recommended for the build.

What This Build Requires Locally

- Flutter SDK + Android Studio/emulator for offline-first mobile development, plus real Android device testing (emulators can't reliably validate offline/sync behaviour)
- Next.js dev server + Node tooling for the web admin/dashboard
- PostGIS-backed local/staging database work for route and territory geo-queries
- Coding agent (bigpickle/OpenCode) running in Zed against NVIDIA NIM cloud models, with Ollama as local fallback
- Multiple concurrent tools — browser, emulator, editor, database client — running side by side

Recommended Upgrades (Dell OptiPlex 9020 baseline)

Upgrade	Approx. Cost (KSh)	Priority
1 TB SATA SSD (replace HDD)	8,000 – 12,000	★★★★★
RAM upgrade to 32 GB DDR3	10,000 – 18,000	★★★★☆
Physical Android test device (mid-range)	12,000 – 25,000	★★★★★
i7-4790 CPU upgrade (if currently i5)	5,000 – 10,000	★★★☆☆
24–27" external monitor	15,000 – 30,000	★★★☆☆
UPS (power backup)	8,000 – 20,000	★★★☆☆

The SSD swap and the physical Android test device are the two highest-impact items — the SSD removes the single biggest bottleneck in daily development speed, and a real device is essential for validating offline capture and sync behaviour in actual field conditions rather than emulator conditions. RAM and CPU upgrades are worthwhile but secondary. Estimated total investment: roughly KSh 30,000–60,000 depending on the current CPU and existing monitor.